NFTALEM ASEFAW

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EDUCATION

University of Maryland, College Park — College Park, MD
Bachelor of Science in Electrical Engineering — Expected Dec 2028

Montgomery College — Rockville, MD
Associate of Science in Engineering Science (Electrical Engineering), GPA: 3.82 — May 2026

Relevant Coursework: Electric Circuits, Digital Logic Design, Digital Circuits and Systems Lab, Discrete Signal Analysis

SKILLS

Programming: C, Verilog, Python (NumPy, PyTorch), MATLAB, JavaScript (Node.js, Express, React), C#
Hardware: Embedded Systems, Raspberry Pi, Arduino, FPGA (Basys 3), Sensor Integration (SPI, I2C), Breadboarding
Lab Equipment: Oscilloscope, Logic Analyzer, Function Generator, DC Power Supply, Multimeter
Simulation and Tools: Vivado, PSpice (OrCAD Capture CIS), EDA Playground, Unity, Git, SQLite, Socket.IO, Excel
Modeling: CREO, Tinkercad, 3D Printing, Logic Aid, Circuit Maker

TECHNICAL EXPERIENCE

Sky Zone Kitchen Display System — Gaithersburg, MD
Lead Engineer — Mar 2026 – Present

- Designed and deployed a real-time kitchen display system that replaced verbal order calls in a busy café kitchen.
- Developed the full stack on a Raspberry Pi server: Node.js, Express, and Socket.IO for live tablet-to-kitchen order sync, SQLite for order history, and a React interface with urgency timers and party-order scheduling.
- Traced tablet connection failures to unreliable Android mDNS and fixed them with a router and DHCP reservations.
- Tripled orders completed in the same time and cut wait times from 10–20 to about 9 minutes; adopted by all kitchen staff.

Neural Network and GAN Projects — College Park, MD
Student Developer — Sep 2026 – Present

- Built a handwritten digit recognizer from scratch in Python and NumPy, comparing 26 network configurations in a parameter study and reaching 97.3% accuracy on 10,000 test images.
- Trained GANs in PyTorch on a Colab GPU, including a face generator trained on 20,000 photos.

RFID Clock-In System — Rockville, MD
Co-Leader — Sep 2025 – Dec 2025

- Co-led the build of a Raspberry Pi time clock that reads RC522 RFID badges over SPI and displays status on an I2C LCD.
- Programmed badge authentication, UID parsing, GPIO control, timestamp logging, and pay calculation in Python.

SeaPerch Underwater Robotics Competition — Biloxi, MS
Robotics Team Lead — Apr 2024 – May 2024

- Directed the build of a tethered underwater ROV, tuning buoyancy, stability, and thruster layout through iterative testing.
- Won 1st at the Southern Regional and placed 32nd of 75 high school teams at the 2024 International SeaPerch Challenge.

LEADERSHIP EXPERIENCE

Northstar Tutoring — Hyattsville, MD
Volunteer Tutor — Oct 2025 – Present

- Mentor two high school students in math and computer science, contributing to a 0.5-point average GPA increase.

Sky Zone Trampoline Park — Gaithersburg, MD
Team Lead, Café Associate, Party Host, Front Desk Cashier — Jul 2024 – Present

- Supervise daily operations across park attractions, coordinating staff and safety procedures during peak hours.
- Onboard new employees on operating procedures, equipment use, and customer safety.

AWARDS AND AFFILIATIONS

Awards: Presidential Scholars Program Scholarship (2025–26), 2nd Place, MC Design Challenge (2025), Dean's List (2024–25)
Affiliations: Center for Minorities in Science and Engineering (CMSE) (2026), Phi Theta Kappa Honor Society (2025–26), IT, Math, and Science Club (2022–24)